



Complex Systems

2022/2023

Mid-semester Exam 29 of March 2023 Time: 2h00

Name:

Number:

This is a **closed books** exam. Any attempt at cheating implies immediate disapproval in the course and the activation of the procedures defined in the disciplinary regulation document of the UC.

Question #	Points	Your score
1	7.5	
2	5	
3	22.5	
4	20	
5	20	
6	12.5	
7	12.5	
	100	

Grader:

Question 1 - Define 7.5 points

Define the following concepts in a clear and **rigorous** way. If applicable, you may use a concrete example to **help you** to explain the concept.

a) Emergence.

Your answer:

b) Fixed-Point.

Your answer:

c) Cobweb.

Your answer:

Question 2 - **Comment** 5 points

Comment the following statements. **Remember**, to comment is not just to say agree/disagree for you have to justify your evaluation.

- a) The dynamical system $\frac{dx}{dt} = \sin(x)$ is an example of a first-order dynamical system with an **oscillatory** behavior.

Your answer:

Your answer:

- b) We only have to study linear, first-order, autonomous systems (LFA) because **all other types** of systems can be converted into a LFA system.

Your answer:

Your answer:

Question 3 - Solve 22.5 points

Consider the following 1D dynamical system:

$$\frac{dx}{dt} = 2x - x^2$$

- a) Find the **fixed-points** of the system;

Your answer:

b) Sketch the corresponding **phase line** (fixed-points and trajectories);

Your answer:

c) Concerning the **stability** what are the characteristics of the fixed-points?

Your answer:

Your answer:

Question 4 - Explain 20 points

In class we discussed the **Euler Method** used to numerically solve differential equations.

- a) Explain how the method works.

Your answer:

Your answer:

b) Simulate the **first two steps** of the method when applied to the system

$$\frac{dx}{dt} = x^2 - 4$$

Your answer:

Your answer:

Question 5 - Model 20 points

Imagine a population in which the organisms may be at three different stages of development: larvae, young adults and adults. Suppose that 1% of larvae grow into young adults every year, and 30% of young adults survive to become (older) adults. Furthermore, young adults have an average of 50 offspring each year, while older adults have an average of 100 offspring each year. Finally, the yearly dead rate of larvae, young adults, and adults is, respectively 3%, 30% and 50%.

- a) Based on these relations model the dynamics using **difference equations**.

Your answer:

Your answer:

- b) Supposing that initially you have only 100 larvae, compute the value of larvae, young adulta and adults **after two years**.

Your answer:

Your answer:

Question 6 - [Model](#) 12.5 points

As you know, some economists say that there is a relationship between the supply of a product and its price. The idea is that, if the quantity of the product is below a certain fixed value (threshold) the price will rise proportionally to the **difference** to this value. The price falls in the same proportion if the quantity is above the threshold. The same happens for the quantity of the product: it must decrease if the price is below a certain fixed value (threshold), increase otherwise. Here also, the increase/decrease must be proportional to the difference between the actual quantity and the threshold. Present, and justify, a **model** of this problem.

Your answer:

Your answer:

Question 7 - Model 12.5 points

In class we presented two different types of predator-prey models, one based on difference equations, and the other on differential equations. Both assume just one predator and one prey. This is not realistic. Suppose that our ecosystem has two competing predators (e.g., lions and tigers) and one prey (e.g., zebras). Present, and justify, a **model** of this new situation.

Your answer:

Your answer: